

agents.md

Autonomous AI Research Assistant and Reproduction Agent

Vision

Build an AI-powered research assistant that functions as a virtual research intern capable of:

- Understanding research papers
- Performing literature reviews
- Discovering related work
- Assisting with learning and comprehension
- Reproducing research experiments
- Diagnosing reproduction failures
- Generating professional reports

The system should automate significant portions of the research workflow while keeping the researcher in control.

High-Level Workflow

Researcher

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Provide Topic / Paper

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Paper Understanding Agent

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Knowledge Base Creation

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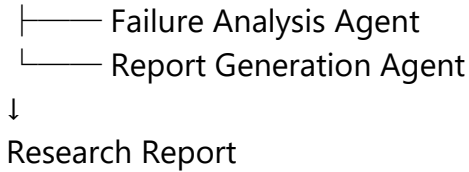
Research Assistant Agents

|—— Literature Discovery Agent

|—— Summarization Agent

|—— Learning Agent

|—— Reproduction Agent



Agent Architecture

1. Orchestrator Agent

Role:

Central coordinator of the entire system.

Responsibilities:

- Receive user requests
- Create execution plans
- Delegates tasks
- Monitor agent execution
- Aggregate results
- Handle failures

Inputs:

- User query
- Topic
- PDF
- arXiv, ScienceDirect link

Outputs:

- Task execution graph
- Agent assignments
- Final aggregated response

Example:

User:

"Study RAG evaluation techniques."

Orchestrator:

1. Find papers.

2. Summarize papers.
 3. Generate learning notes.
 4. Produce report.
-

2. Paper Understanding Agent

Purpose

Convert research papers into machine-understandable knowledge.

Responsibilities

- Parse PDF
- Extract sections
- Extract metadata
- Identify contributions
- Extract methodology
- Extract results

Tools

- PyMuPDF (to process pdfs)
- Unstructured (better choice than PyMuPDF)
- LLM

Inputs

- PDF
- arXiv paper

Outputs

```
{  
  metadata,  
  sections,  
  contributions,  
  findings  
}
```

3. Knowledge Base Agent (to be combined w 2)

Purpose

Create searchable research memory.

Responsibilities

- Chunk paper
- Generate embeddings
- Store vectors
- Maintain metadata

Tools

- ChromaDB
- BGE Embeddings

Outputs

Searchable research corpus. Basically forming the search space for RAG.

4. Research Chat Agent (to be combined w 2)

Purpose

Allow conversational interaction with papers.

Responsibilities

- Retrieve context
- Generate grounded answers
- Cite sources
- Handle follow-up questions

Tools

- Retriever
- LLM

Example

User:

"What is the main contribution?"

Agent:

Returns contribution with citations.

5. Literature Discovery Agent(API + Web scraping)

Purpose

Perform literature review.

Responsibilities

- Find related papers
- Find foundational work
- Find citations

Sources

- Semantic Scholar
- arXiv
- OpenAlex

Outputs

Related papers.

6. Summarization Agent (Initially with 2, later distinction)

Purpose

Condense large amounts of information.

Responsibilities

- Generate abstracts

- Generate executive summaries
- Extract key points
- Generate highlights

Outputs

Short Summary

Technical Summary

Research Summary

7. Learning Agent (Phase 2)

Purpose

Act as a study companion.

Responsibilities

- Generate notes
- Create flashcards
- Create quizzes
- Explain concepts

Outputs

Study Material

Quiz Bank

Flashcards

Interview Questions

8. Research Gap Agent (Optional)

Purpose

Identify unexplored research opportunities.

Responsibilities

- Compare papers
- Identify limitations
- Detect open problems
- Suggest future directions

Outputs

Research Gap Report

Potential Project Ideas

Future Research Suggestions

9. Reproduction Agent (Best with Claude Code)

Purpose

Reproduce published research.

Responsibilities

- Locate GitHub repositories
- Locate datasets
- Extract requirements
- Build execution environment
- Execute experiments

Subtasks

Repository Discovery

Dataset Discovery

Environment Setup

Execution

Evaluation

Outputs

Reproduction Status

10. Environment Setup Agent (Phase 2)

Purpose

Prepare isolated execution environments.

Responsibilities

- Parse requirements.txt
- Parse environment.yml
- Build Docker containers
- Resolve dependencies

Outputs

Docker Image

Dependency Graph

Setup Logs

11. Execution Agent (Phase 2)

Purpose

Run experiments.

Responsibilities

- Execute code
- Capture outputs
- Capture metrics
- Monitor resources

Outputs

Runtime Metrics

Execution Logs

Model Outputs

12. Failure Analysis Agent(Phase 2)

Purpose

Diagnose experiment failures.

Responsibilities

- Parse logs
- Categorize failures
- Suggest fixes
- Recommend retries

Common Errors

CUDA Mismatch

Missing Dataset

Dependency Conflict

Invalid Paths

Network Errors

Outputs

Failure Report

Recommended Fixes

Confidence Score

13. Report Generation Agent

Purpose

Produce professional reports.

Responsibilities

- Aggregate outputs
- Format findings
- Create markdown reports

Report format:

- Introduction
- Objectives
- Methodology
- Requirements
- Implementation
- Results
- References
- HTML

PDF (Future)

Execution Flow

Workflow A: Understanding a Paper

User Uploads PDF

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Paper Understanding Agent

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Knowledge Base Agent

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Research Chat Agent

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Summary Agent

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Learning Agent (If needed and when integrated)

Deliverables:

- Summary
 - Notes
 - Quiz
 - Chat Interface
-

Workflow B: Literature Review

Research Topic

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Literature Discovery Agent

↓

Summarization Agent (later, right now we only have the links to similar papers)

Deliverables:

- Related Papers
-

Workflow C: Reproduction

Paper

↓

Repository Discovery

↓

Dataset Discovery

↓

Environment Setup

↓

Execution Agent

↓

Failure Analysis

↓

Report Generation

Deliverables:

- Reproduction Checklist
 - Execution Logs
 - Failure Report
 - Reproducibility Score
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Deliverables

MVP

- PDF Upload

- arXiv Integration
- RAG Chat
- Paper Summary
- Related Paper Search

Version 2

- GitHub Discovery
 - Dataset Discovery
 - Reproduction Checklist
-

Version 3

- Docker Execution
 - Autonomous Experiment Running
 - Failure Diagnosis
-

Recommended Project Structure

backend/

|—— api/

|—— agents/

| |—— orchestrator/

| |—— paper_agent/

| |—— literature_agent/

| |—— learning_agent/

| |—— reproduction_agent/

| |—— failure_agent/

| |—— report_agent/

|

|—— rag/

|—— vector_store/

|—— tools/

|—— workflows/

|—— database/

|—— reports/

frontend/

|—— dashboard/

|—— paper_viewer/

|—— chat/

|—— reproduction/

|—— reports/

infra/

|—— docker/

|—— monitoring/

|—— deployment/

Recommended Tech Stack

Frontend

- React.js
- TailwindCSS
- shadcn/ui

Backend

- Python
- FastAPI

Agents

- LangGraph

RAG

- ChromaDB
- BGE Embeddings

LLMs

- GPT-5
- Qwen3
- Llama 4

Research APIs

- Semantic Scholar
- OpenAlex
- arXiv

Execution

- Docker
- Docker Compose

Storage

- PostgreSQL

Observability

- LangSmith
- OpenTelemetry

Deployment

- Docker
- AWS / Azure / GCP